

ibm360_ipl_sop_v1json

```
{
  "id": "SOP-IBM360-001",
  "title": "IBM System/360 Model 40 IPL Failure Diagnostic",
  "system": "IBM System/360 Model 40",
  "component": "2040 Processing Unit / I/O Channels",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM Field Engineering Maintenance Manual (SY22-2841)",
  "description": "Diagnostic procedure for when Initial Program Load (IPL) fails to complete or
results in a hard wait state.",
  "steps": [
    {
      "step": 1,
      "action": "Console State Verification",
      "details": "Check the System Control Panel. Verify the 'Wait' light status and the
'Load' light status.",
      "indicators": {
        "Wait light ON": "The CPU has entered a wait state. Check the PSW (Program Status
Word).",
        "Load light remains ON": "IPL sequence is stuck. Check the I/O channel and device
status."
      }
    },
    {
      "step": 2,
      "action": "PSW Analysis",
      "details": "Use the console switches to display the current PSW. Check the System Mask
and Instruction Address.",
      "checks": [
        "Bits 14 and 15 (Wait and Problem state bits).",
        "Instruction Address 000-007 (First 8 bytes loaded during IPL)."
      ]
    },
    {
      "step": 3,
      "action": "I/O Error Check",
      "details": "Examine the CSW (Channel Status Word) at real address 64 (decimal).",
      "status_flags": {
        "Unit Check": "The I/O device (e.g., 2311 Disk or 2400 Tape) reported a hardware
error.",
        "Channel Data Check": "A parity error occurred during the transfer of the IPL
record."
      }
    },
    {
      "step": 4,
      "action": "Sense Operation",
      "details": "If 'Unit Check' is present in the CSW, perform a Sense command to the
device to retrieve detailed error bytes.",
      "logic": "If Byte 0 Bit 0 (Command Reject), the IPL record format is invalid."
    }
  ],
  "diagnostic flowchart": "

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{
  "condition": "Wait Light ON immediately after Load",
  "next": "IPL record not found or invalid. Check address switches."
},
{
  "condition": "Machine Check light ON",
  "next": "CPU hardware failure. Run FLT (Fault Location Tests)."
},
{
  "condition": "IPL stops at specific address",
  "next": "Check the IPL program code or device readiness."
}
]
}
```